

Nagastra 1: *First Indigenous Loiter Munition with abort, recover and reuse capability, under production at Solar Industries at Nagpur*

SOLAR INDUSTRIES: RISING TO DEFEND INDIA, WHATEVER BE THE CHALLENGE.

An explosives company led by a visionary patriot has taken upon itself the task of defending the nation. With changing times, Solar Industries has evolved to produce contemporary war craft products that enable our nation to neutralize threats that emanate across dimensions.

Founded by Satyanarayan Nandlal Nuwal in 1996, Solar Group of Industries is today spread across 2600 acres, employing over 8000 people. As Chairman of Solar Industries India Ltd. and its subsidiary Economic Explosives India Ltd. (EEL), his visionary leadership has guided the Solar Group to evolve from a single site manufacturing company into a globally reputed industry dealing with industrial explosives, military explosives, propellants, detonators, boosters, and pyro devices. Today the group has manufacturing facilities across 4 continents meeting the demands of over 60 countries.

Solar has one of the world's largest HMX state-of-the-art production plant at Nagpur to manufacture HMX and HMX based compositions, which were imported. Post meeting India's requirements, it regularly exports HMX and its compositions to countries like the USA, France, Germany, Israel and other European countries.

EEL is the first private company in India to produce complete ammunition. It has developed and supplied 10 Lakhs grenades with a reliability of 99.86%, replacing the vintage M36 grenade. It is also negotiating with many friendly countries to export this grenade.

The group has facilities for development and manufacturing of 30 mm ammunition, anti-tank mines, warheads, missiles, air bombs, etc. With technology support from DRDO, EEL has developed 3 variants of the Pinaka Multi-barrel Rocket System. There is a huge requirement of Pinaka Rocket Systems from our armed forces and EEL has augmented its production infrastructure to meet these requirements. There is also an export order from Armenia.

The company is developing the Maheshvarastra Long Range Precision Rocket System to enhance the indigenous artillery strike capability up to 300 km. The system will be INS/GPS guided to precisely hit the targets with CEP of less than 10 m having more than 75% indigenous contents.

Development and manufacturing of loitering munitions with different endurances, ranges and warheads are at the forefront of the weapons being developed by various countries. Solar Industries is developing the Nagastra series of Loiter Munitions of ranges from 15 km to more than 100 km and payload carrying capability from 1 kg to 10 kg. These Loiter Munitions can be called back and reused in case the target is not found or the mission is aborted. After successful user's trials Nagastra 1 has gone into production and Nagastra 2 & Nagastra 3 are under advanced stage of development.

To counter drones deployed by adversaries, EEL is developing the Bhargavastra Counter Drone System, which has the capability of inflicting hard kill against swarms of enemy drones using micro guided missiles.

Solar Group has created facilities for the development of ammunition and defence systems. State of art test facilities for performance evaluation of rockets, warheads/ ammunition and environmental facilities to qualify the explosive systems under all weather conditions exist in its premises at Nagpur. It is the first to develop a World Class Facility in the country to determine the Shelf life of Ammunition and Energetic Material based on Isothermal MicroCalorimetry (IMC) , revolutionizing the existing 70 years old primitive ISAT based procedure of shelf life assessment.

It is the passion of Satyanarayan Nuwal to make our country self-reliant in ammunition and defence systems. Under his leadership the Solar group is committed to realize the government mandate of Make in India and Make for the World. The group has ventured into R&D, technology refinement, new product development and life cycle management of high energy materials. It has built a competent team of motivated human resources and established alliances & partnerships with other industries & start-ups to develop state of the art defence systems with high indigenous contents and meeting global quality standards.

Satyanarayan Nuwal's business acumen has ensured a consistently high growth trajectory for the group's revenues. Solar Industries went public in 2006. Listed in both BSE & NSE, its market capitalization as of now is about Rs. 62,000 Crores.